

Grape Resonances

Data Appendix

Corpus Listings, Curated Data, and Source Materials

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Companion documents: Summary · Methods Primer · Technical Appendix · Data Appendix · Region Reference (Identity)

This appendix presents the data the framework reads against: the 59 region corpus inherited from Region Resonances, the 101 canonical grape surface specific to Grape Resonances, the 12-entry alias map, the 11 silent-skipped non-canonical grapes, the 10 framing sentences in full, and the per-region signature curation produced across four sessions in May 2026. It also documents the curation methodology and source materials. The data files described here are static, locked at build time, and shipped as JSON with the deployed tool.

A. The Region Corpus

A.1 The 59 Regions

The corpus is identical to Region Resonances, comprising 59 wine regions partitioned into the six Soul of Wine clusters established in Region Affinities. The table below gives each region's cluster, the one-word metaphor that opens its identity narrative, and the country (where the region is sub-national) or canonical regional identifier.

Region	Cluster	Metaphor	Country/Area
Burgundy	Old World Interior	Devotion	France
Northern Rhône	Old World Interior	Solitude	France
Mosel	Old World Interior	Poetry	Germany
Rheingau	Old World Interior	Continuity	Germany
Tokaj	Old World Interior	Melancholy	Hungary
Campania	Old World Interior	Memory	Italy
Etna	Old World Interior	Revival	Italy
Piedmont	Old World Interior	Severity	Italy
Galicia	Old World Interior	Morriña	Spain
Santa Cruz Mountains	Old World Interior	Cost	USA (CA)
Beaujolais	Outward Ease	Joy	France
Provence	Outward Ease	Tranquility	France
Veneto	Outward Ease	Commerce	Italy
Marlborough	Outward Ease	Assertion	New Zealand
Walla Walla	Outward Ease	Community	USA (WA)
Dalmatian Coast	Old World Exterior	Tranquility	Croatia
Bordeaux	Old World Exterior	Business	France
Champagne	Old World Exterior	Society	France
Châteauneuf-du-Pape	Old World Exterior	Generosity	France
Baden	Old World Exterior	Warmth	Germany
Pfalz	Old World Exterior	Generosity	Germany

Region	Cluster	Metaphor	Country / Area
Rioja	Old World Exterior	Patience	Spain
Wagram	Against the Odds	Earth	Austria
Jura	Against the Odds	Eccentricity	France
Macedonia	Against the Odds	Austerity	Greece
Santorini	Against the Odds	Survival	Greece
Liguria	Against the Odds	Intimacy	Italy
Sardinia	Against the Odds	Stubbornness	Italy
Douro	Against the Odds	Endurance	Portugal
Ribera del Duero	Against the Odds	Severity	Spain
Barossa Valley	Against the Odds	Fortitude	Australia
Hunter Valley	Against the Odds	Defiance	Australia
Sicily	New World Reinvention	Resurrection	Italy
Catalonia	New World Reinvention	Rebellion	Spain
Mendoza	New World Reinvention	Acceleration	Argentina
Patagonia	New World Reinvention	Extremity	Argentina
Central Otago	New World Reinvention	Adventure	New Zealand
Stellenbosch	New World Reinvention	Aspiration	South Africa
Swartland	New World Reinvention	Rebellion	South Africa
Columbia Valley	New World Reinvention	Ambition	USA (WA)
Finger Lakes	New World Reinvention	Conviction	USA (NY)
Napa Valley	New World Reinvention	Ambition	USA (CA)
Paso Robles	New World Reinvention	Independence	USA (CA)
Sonoma	New World Reinvention	Authenticity	USA (CA)
Kamptal	The Moderates	Discipline	Austria
Steiermark	The Moderates	Dialogue	Austria
Wachau	The Moderates	Monumentality	Austria
Alsace	The Moderates	Duality	France
Loire	The Moderates	Sentimentality	France
Nahe	The Moderates	Subtlety	Germany
Alto Adige	The Moderates	Precision	Italy
Friuli-Venezia Giulia	The Moderates	Clarity	Italy
Tuscany	The Moderates	Art	Italy
Goriška Brda	The Moderates	Fortune	Slovenia
Margaret River	The Moderates	Composure	Australia
Maipo Valley	The Moderates	Pride	Chile
Hawke's Bay	The Moderates	Confidence	New Zealand
Santa Barbara	The Moderates	Serendipity	USA (CA)
Willamette Valley	The Moderates	Idealism	USA (Oregon)

A.2 The Embedding Model and Operating Characteristics

The embedding model is OpenAI’s `text-embedding-3-small`, the same model used by Region Resonances and used unchanged in Grape Resonances. It produces unit-normalised 1536-dimensional vectors. The 59 region passages (the metaphor concatenated with the full Layer 1 identity narrative) are embedded once at build time; the resulting matrix $R \in \mathbb{R}^{59 \times 1536}$ is shipped with the tool as `region_embeddings.json`.

The empirical operating range of cosine similarity on this corpus is approximately $[0.15, 0.65]$ for natural-language English queries, narrowing toward $[0.15, 0.45]$ for typical short-phrase queries and broadening toward $[0.55, 0.70]$ when the query is itself a paraphrase of source narrative. Values outside these ranges occur only for pathological inputs. The full operating-characteristic discussion appears in the Region Resonances Data Appendix; the matcher-stage display behaviour

is unchanged in Grape Resonances.

B. The Canonical 101-Grape Surface

The canonical grape surface $\mathcal{C}$ is the set of grape names that the response pool is closed to. It contains 101 entries partitioned into 53 reds and 48 whites. The surface is locked at build time. Eleven grapes that appear in regional signature curations but are not on the canonical surface are documented in §D.

B.1 Red Grapes (53)

Aglianico · Baga · Barbera · Blaufränkisch · Bobal · Cabernet Franc · Cabernet Sauvignon · Carignan · Carménère · Cinsault · Corvina · Cunoise · Dolcetto · Frappato · Gamay · Garnacha · Tintorera · Graciano · Grenache · Kadarka · Lagrein · Limniona · Malbec · Mavrud · Mencía · Merlot · Monastrell · Montepulciano · Mourvèdre · Nebbiolo · Negroamaro · Nerello Mascalese · Nero d’Avola · Petit Verdot · Petite Sirah · Pinot Noir · Pinotage · Plavac Mali · Poulsard · Primitivo · Sagrantino · Sangiovese · Schiava · St. Laurent · Syrah · Tannat · Tempranillo · Tinta Roriz · Touriga Franca · Touriga Nacional · Trousseau · Xinomavro · Zinfandel · Zweigelt.

B.2 White Grapes (48)

Albariño · Aligoté · Arinto · Arneis · Assyrtiko · Chardonnay · Chasselas · Chenin Blanc · Clairette · Colombard · Cortese · Encruzado · Falanghina · Fiano · Friulano · Furmint · Garganega · Gewürztraminer · Godello · Greco · Grüner Veltliner · Loureiro · Malagousia · Malvasia · Marsanne · Mauzac · Melon de Bourgogne · Müller-Thurgau · Muscat Blanc · Petit Manseng · Pinot Blanc · Pinot Gris · Ribolla Gialla · Riesling · Roussanne · Sauvignon Blanc · Silvaner · Semillon · Torrontés · Trebbiano · Ugni Blanc · Verdejo · Verdelho · Verdicchio · Vermentino · Vidiano · Viognier · Welschriesling.

B.3 Per-Grape Narratives

Each canonical grape has a one- to two-sentence character note in Soul of Wine voice, stored in `grape_narratives.json`. The notes are character-focused rather than tasting-note-focused: they describe what the grape *is* in the project’s framework rather than what its wines taste like. The notes were drafted under a methodology that prohibited cross-referencing against the existing notes, to prevent voice drift, and were reviewed for stylistic recurrences after completion. They are not reproduced in full here because of their volume; they surface in the trace fold of every Grape Resonances response.

C. The Alias Map

The alias file `grape_aliases.json` contains 12 entries mapping a regional spelling to its canonical name. Each entry records a one-to-one botanical correspondence.

Regional Spelling	Canonical Name
Spätburgunder	Pinot Noir

Regional Spelling	Canonical Name
Shiraz	Syrah
Pinot Grigio	Pinot Gris
Pinot Bianco	Pinot Blanc
Weissburgunder	Pinot Blanc
Grauburgunder	Pinot Gris
Tinto Fino	Tempranillo
Cannonau	Grenache
Garnatxa (Grenache)	Grenache
Carinyena (Carignan)	Carignan
Carignano	Carignan
Semillon (Hunter Valley)	Sémillon

The alias resolution preserves the regional spelling in the disclosed trace. A response that includes Syrah, drawn from Walla Walla and Barossa Valley, shows the trace as *Walla Walla* — *community* and *Barossa Valley* — *fortitude* · *as Shiraz*. The user sees both that the alias resolution occurred and what the regional spelling was.

D. The Silent-Skip List

Eleven signature grapes that appear in regional curations are neither canonical (in the 101-grape surface) nor aliased to a canonical name. These are silent-skipped from the response pool by the aggregator: the region still contributes whatever canonical or aliased grapes it has, but the non-canonical grape is not counted toward any recurrence. The list is documented and may expand in a future version of the canonical surface.

Grape	Primary Region(s) of Signature Appearance
Carricante	Etna (signature; partner to Nerello Mascalese)
Glera	Veneto (signature; Prosecco’s base variety)
Greco di Tufo	Campania (signature; DOCG variety)
Grillo	Sicily (signature; Marsala and modern dry whites)
Macabeu	Catalonia (signature; Cava base variety)
Pigato	Liguria (signature; coastal Ligurian white)
Pinot Meunier	Champagne (signature; the third of the three Champagne grapes)
Pošip	Dalmatian Coast (signature; Korčula island white)
Rebula (Ribolla Gialla)	Goriška Brda (signature; the Slovenian name for Ribolla Gialla)
Rossese	Liguria (signature; Rossese di Dolceacqua)
Savagnin	Jura (signature; vin jaune base variety)

Note that Rebula appears here despite Ribolla Gialla being on the canonical surface. The alias file does not contain a Rebula → Ribolla Gialla mapping in the v1 release; the two names are treated as separate strings at the aggregator’s lookup stage. This is a known v1 limitation; the planned addition to the alias file would resolve it in v1.1.

E. The Framing Sentences

Ten one-sentence framings, hand-written in Soul of Wine voice, are stored in `cluster_framings.json`. The coherence detector’s verdict (*bullseye for cluster k* or *prism*) selects one of these as the

framing line beneath the grape names in the response.

E.1 Bullseye Framings (one per cluster)

Old World Interior. Your word points to the patient, custodial cultures — places where time is the essential ingredient and the winemaker is a keeper, not a maker.

Outward Ease. Your word points to the communal cultures — places where wine is part of the table, the market, the gathering, never separate from the company it keeps.

Old World Exterior. Your word points to the cultures of inheritance — places where wine is part of how the society organises itself: through family, through trade, through patience, through the long table.

Against the Odds. Your word points to the cultures of endurance — places where difficulty is not an obstacle but the substance.

New World Reinvention. Your word points to the cultures of conviction-without-evidence — places that decided they should exist and then made themselves.

The Moderates. Your word points to the cultures of found voice — places that arrived at their own centre, each by a different route, none by choosing the loudest register.

E.2 Prism Framings

General prism. Your word refracts. In this framework's reading, it is not one feeling but several — the regions your phrase pulls belong to different temperaments, each holding a facet of what you brought.

Interior × Reinvention prism (Old World Interior crossed with New World Reinvention). Your word holds a tension between patient inheritance and chosen invention — wine cultures that arrived at depth by very different routes.

Difficulty × Ease prism (Against the Odds crossed with Outward Ease). Your word holds a tension between difficulty and ease — wine cultures that found themselves under pressure beside cultures that found themselves at the table.

Interior × Moderate prism (Old World Interior crossed with The Moderates). Your word holds a tension between the cultures of long depth and the cultures of found equilibrium — different settlements of the same question about how to make wine seriously.

F. Per-Region Signature Curation

For each of the 59 regions, the curated file `region_grapes.json` associates the region with three tiers of grape varieties (*signature*, *lesser known*, *rare*) and a derived field (*temperament seconds*) drawn from the lesser-known tier. The signature tier is the only tier the v1 aggregator reads. The full per-region signature lists are given below, with the temperament-seconds field (a single grape per region, possibly empty) in the rightmost column.

Region	Signature Grapes	Temperament Seconds
<i>Old World Interior</i>		
Burgundy	Pinot Noir, Chardonnay	Aligoté
Northern Rhône	Syrah, Viognier	Marsanne
Mosel	Riesling	Elbling
Rheingau	Riesling	Spätburgunder
Tokaj	Furmint	Hárslevelű
Campania	Aglianico, Fiano, Greco di Tufo	Piediroso
Etna	Nerello Mascalese, Carricante	Nerello Cappuccio
Piedmont	Nebbiolo	Freisa
Galicia	Albariño, Mencía, Godello	Treixadura
Santa Cruz Mountains	Pinot Noir, Cabernet Sauvignon	Zinfandel
<i>Outward Ease</i>		
Beaujolais	Gamay	(empty)
Provence	Grenache	Cinsault
Veneto	Glera, Garganega, Corvina	Pinot Grigio
Marlborough	Sauvignon Blanc	(empty)
Walla Walla	Cabernet Sauvignon, Syrah	Merlot
<i>Old World Exterior</i>		
Dalmatian Coast	Plavac Mali, Pošip	Babić
Bordeaux	Cabernet Sauvignon, Merlot, Cabernet Franc	Sémillon
Champagne	Pinot Noir, Chardonnay, Pinot Meunier	Pinot Gris (Fromenteau)
Châteauneuf-du-Pape	Grenache, Syrah, Mourvèdre	Counoise
Baden	Spätburgunder, Grauburgunder, Weissburgunder	Gutedel
Pfalz	Riesling, Spätburgunder	Dornfelder
Rioja	Tempranillo	Graciano
<i>Against the Odds</i>		
Wagram	Grüner Veltliner	Roter Veltliner
Jura	Savagnin, Chardonnay, Poulsard	Trousseau
Macedonia	Xinomavro	Assyrtiko
Santorini	Assyrtiko	Aidani
Liguria	Pigato, Vermentino, Rossese	Bosco
Sardinia	Cannonau, Vermentino, Carignano	Vernaccia di Oristano
Douro	Touriga Nacional, Touriga Franca, Tinta Roriz	Tinto Cão
Ribera del Duero	Tinto Fino	Albillo
Barossa Valley	Shiraz	Grenache
Hunter Valley	Semillon (Hunter Valley), Shiraz	Verdelho
<i>New World Reinvention</i>		
Sicily	Nero d'Avola, Grillo	Perricone
Catalonia	Garnatxa (Grenache), Carinyena (Carignan), Macabeu	Xarel·lo
Mendoza	Malbec, Cabernet Sauvignon	Cabernet Franc
Patagonia	Pinot Noir, Malbec	Sémillon
Central Otago	Pinot Noir	(empty)
Stellenbosch	Cabernet Sauvignon, Chenin Blanc	Cabernet Franc
Swartland	Chenin Blanc, Syrah	Cinsault
Columbia Valley	Cabernet Sauvignon, Merlot, Syrah	Riesling
Finger Lakes	Riesling	Cabernet Franc
Napa Valley	Cabernet Sauvignon, Merlot	Zinfandel
Paso Robles	Zinfandel, Cabernet Sauvignon, Syrah	Grenache

Region	Signature Grapes	Temperament Seconds
Sonoma	Pinot Noir, Chardonnay, Zinfandel, Cabernet Sauvignon	Carignan
<i>The Moderates</i>		
Kamptal	Grüner Veltliner, Riesling	Zweigelt
Steiermark	Sauvignon Blanc, Welschriesling	Gelber Muskateller
Wachau	Riesling, Grüner Veltliner	(empty)
Alsace	Riesling, Gewürztraminer, Pinot Gris	Muscat (Blanc à Petits Grains + Ottonel)
Loire	Chenin Blanc, Sauvignon Blanc, Cabernet Franc	Pineau d'Aunis
Nahe	Riesling	Silvaner
Alto Adige	Pinot Grigio, Gewürztraminer, Pinot Bianco	Kerner
Friuli-Venezia Giulia	Friulano, Pinot Grigio	Ribolla Gialla
Tuscany	Sangiovese, Cabernet Sauvignon, Merlot	Canaiolo
Goriška Brda	Rebula (Ribolla Gialla), Merlot	Pinela
Margaret River	Cabernet Sauvignon, Chardonnay	Semillon
Maipo Valley	Cabernet Sauvignon	Carménère
Hawke's Bay	Cabernet Sauvignon, Merlot, Chardonnay	Syrah
Santa Barbara	Pinot Noir, Chardonnay	Syrah
Willamette Valley	Pinot Noir	Gamay Noir

The full per-region tier data, including the *lesser known* and *rare* entries (which the v1 aggregator does not read), is shipped as `region_grapes.json` with the deployed tool and is available in the source repository.

G. Curation Methodology

G.1 Tier Definitions

The curation is governed by four field definitions, locked in the `_metadata.curation_rules` block of `region_grapes.json`:

- *Signature* — what the region is famous for; 1–4 grapes (4 used only when the regional identity is genuinely multi-pillar, e.g. Sonoma, Alsace). The v1 aggregator reads from this tier exclusively.
- *Lesser known* — recognised by occasional readers but not headline; the tier from which *temperament seconds* is drawn.
- *Rare* — even moderately serious enthusiasts haven't heard; up to 4 entries; may be empty.
- *Temperament seconds* — the lesser-known grape that most embodies the region's identity. Always a subset of the lesser-known tier. May be empty for regions without natural Tier-2 candidates (Beaujolais, Marlborough, Central Otago, Wachau).

G.2 Curatorial Principles

Six principles were applied uniformly across all 59 regions:

1. *Region-relative tiers, not market-relative.* Furmint is a signature grape in Tokaj despite being globally lesser-known; Pinot Grigio is a lesser-known grape in Veneto despite global ubiquity. The tier reflects the grape’s position *within the region’s identity*, not its global standing.
2. *Do not retreat from the unfashionable signal.* When a region’s narrative honours commerce, survival, or rebellion, pick the grape that carries that signal even if it is unprestigious. Pinot Grigio for Veneto’s commerce, Sémillon for Bordeaux’s business, Dornfelder for Pfalz’s generosity, Zinfandel for Napa’s ambition, Carignan for Sonoma’s authenticity — all examples of the principle.
3. *Prefer region-specific over cluster-shared.* When two regions could plausibly share a temperament-seconds grape, pick the more region-specific answer for each. Same-grape repeats occur only when the role is genuinely distinct in each region.
4. *The grape with the worst commercial case often wins.* Particularly in Against the Odds and New World Reinvention, where the cluster signals favour endurance and rescue over commercial reasonableness.
5. *Empty is honest.* When a region is genuinely mono-grape or two-grape in its identity (Beaujolais’s Gamay, Marlborough’s Sauvignon Blanc, Wachau’s Riesling-and-Grüner pair, Central Otago’s Pinot Noir), the empty temperament-seconds field is more truthful than a forced second choice.
6. *Cross-border continuums deserve recognition.* Where sibling regions share signatures across borders (Friuli/Brda, Steiermark/Slovenian Štajerska, Bordeaux/Spanish heritage), the temperament-seconds choice differentiates rather than mirrors, preserving the regional specificity within the shared variety pool.

G.3 Same-Grape Cross-Region Patterns

Five grape identities recur as temperament-seconds across two or more regions. Each has a distinct role in each region; the aggregator handles cross-region recurrence through the alias resolution and the role-distinct curation principle. The Sémillon row is split below to reflect the intentional regional spelling distinction in the curation: Sémillon (with accent) is used for French and Spanish-speaking regions; Semillon (no accent) is used for Australian regions per local convention. The two spellings are treated as separate strings in the temperament-seconds field, by curatorial design; the aggregator’s alias map normalises spellings only at the signature-tier lookup, where they need to count as the same grape for recurrence.

Grape	Count	Regions (with role)
Cabernet Franc	3	Mendoza (acceleration), Stellenbosch (aspiration), Finger Lakes (conviction)
Sémillon	2	Bordeaux (generosity), Patagonia (extremity-survival)
Semillon	1	Margaret River (composure) — separate spelling, same variety
Zinfandel	2	Santa Cruz Mountains (patient continuation), Napa Valley (quiet survival under ambition)

Grape	Count	Regions (with role)
Cinsault	2	Provence (tranquility), Swartland (rebellion)
Grenache	2	Barossa Valley (pre-Shiraz survival), Paso Robles (Rhône-Ranger independence)
Syrah	2	Hawke's Bay (quiet confidence), Santa Barbara (serendipitous discovery)

G.4 Validation

The completed curation was validated on the following criteria, all passing across all 59 regions:

- *Subset rule.* Every temperament-seconds entry is contained in the corresponding lesser-known tier.
- *Non-empty signatures.* Every region has at least one signature grape.
- *Non-empty lesser-known.* Every region has at least one lesser-known grape.
- *Empty temperament-seconds documented.* The four regions with empty temperament-seconds (Beaujolais, Marlborough, Central Otago, Wachau) are all single-grape or two-grape regional assertions where the signature itself is the identity.

The curation closed across four sessions over a one-day period (2026-05-10), with one mid-session subset-rule fix (Provence: Cinsault moved from signature to lesser-known to permit Grenache as sole signature) and one mid-session correction (Wachau: Grüner Veltliner restored to signature alongside Riesling).

H. Data File Inventory

The five data files loaded by the deployed tool are:

- **region_embeddings.json** — the matcher's pre-computed region vectors; 1.86 MB; loaded once on app start. Shared with Region Resonances.
- **region_grapes.json** — the per-region tiered grape curation; 30 KB; the substrate for the aggregator.
- **grape_narratives.json** — the 101 per-grape character notes; 30 KB; surfaces in the trace fold.
- **cluster_framings.json** — the 10 framing sentences; 4 KB; surfaces below the grape names in the response.
- **grape_aliases.json** — the 12-entry alias map; 1 KB; consumed by the aggregator before tallying.

All files are static, locked at build time, and shipped with the React build. None is regenerated, modified, or re-tuned at runtime.

I. References

- d’Agata, I. (2014). *Native Wine Grapes of Italy*. University of California Press.
- Galet, P. (2000). *Dictionnaire encyclopédique des cépages*. Hachette.
- Robinson, J. & Harding, J. (eds.) (2015). *The Oxford Companion to Wine*, 4th ed. Oxford University Press.
- Robinson, J., Harding, J. & Vouillamoz, J. (2012). *Wine Grapes: A Complete Guide to 1,368 Vine Varieties*. Allen Lane. Primary reference for canonical grape identities, alias relationships, and the silent-skip determinations.
- OpenAI (2024). New embedding models and API updates. Technical announcement covering `text-embedding-3-small` and `text-embedding-3-large`.